

CO-1

1. Define Artificial Intelligence.

Artificial Intelligence (AI) is a branch of computer science that deals with creating intelligent machines capable of performing tasks that normally require human intelligence.

These tasks include:

- Learning
- Reasoning
- Problem solving
- Decision making
- Language understanding

AI systems simulate human thinking and behavior to solve real-world problems efficiently.

Examples:

- Chatbots
 - Self-driving cars
 - Voice assistants
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2. Define Rationality.

Rationality refers to the ability of an agent to choose the best possible action based on available information and percepts.

A rational agent always tries to maximize the performance measure and achieve the best outcome.

Rationality depends on:

- Performance measure
- Agent's knowledge
- Available actions
- Percept sequence

Example:

A GPS selecting the shortest route is acting rationally.

3. Define Agent.

An agent is an entity that perceives its environment through sensors and acts upon the environment using actuators.

Formula:

Agent = Sensors + Actuators

Examples:

Agent	Sensors	Actuators
Human	Eyes, ears	Hands, legs
Robot	Camera, microphone	Wheels, arms

An intelligent agent selects actions that help achieve goals successfully.

4. Outline Agent Program.

An agent program is a software implementation that maps percepts received from sensors to actions performed by actuators.

It decides what action should be taken for a given percept.

Structure:

Percepts → Agent Program → Actions

Functions:

- Receives input from environment
- Processes information
- Produces suitable action

Example:

A vacuum cleaner agent cleaning dirty rooms automatically.

5. List out the steps in Problem-Solving Process.

The problem-solving process in AI involves the following steps:

1. Goal Formulation
Identify the objective to be achieved.
2. Problem Formulation
Define states, actions, and goals.
3. Search Process
Explore possible solutions.
4. Execute Solution
Apply selected actions.
5. Evaluate Result
Check whether goal is achieved.

Example:

Finding shortest route in map navigation.

6. Compare Path Cost and Step Cost.

Feature	Step Cost	Path Cost
Definition	Cost of one action	Total cost from start to goal
Scope	Single step	Entire path
Calculation	Immediate cost	Sum of all step costs
Example	Cost between two cities	Total travel expense

Conclusion:

Path cost is the sum of all step costs in a solution path.

7. Comment the usage of Evaluation Function.

An evaluation function is used to estimate the desirability or quality of a state during search.

It helps AI systems:

- Choose best moves
- Compare alternatives
- Improve decision making

Representation:

$f(n)$

Applications:

- A* Search
- Minimax Algorithm
- Game Playing AI

Example:

In chess, evaluation function estimates board strength.

8. What is the Branching Factor?

Branching Factor is the number of child nodes generated from a parent node in a search tree.

Represented by:

b

Example:

If each node generates 3 children:

$b = 3$

Importance:

Branching factor affects:

- Time complexity
 - Memory usage
 - Search efficiency
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9. Define Heuristic Function.

A heuristic function is a function that estimates the cost from the current node to the goal node.

Represented as:

$h(n)$

Where:

- $h(n)$ = estimated cost to reach goal

Usage:

Used in informed search algorithms such as:

- A* Search
- Greedy Best First Search

Advantage:

Reduces search time by guiding search intelligently.

10. List out the types of Uninformed Search Strategies.

Uninformed search strategies search without heuristic information.

Types:

1. Breadth First Search (BFS)
2. Depth First Search (DFS)
3. Uniform Cost Search
4. Depth Limited Search
5. Iterative Deepening Search
6. Bidirectional Search

Features:

- Uses only problem definition
 - Blind search technique
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11. Define Iterative Deepening Search.

Iterative Deepening Search (IDS) repeatedly applies Depth First Search with increasing depth limits until the goal is found.

It combines advantages of:

- BFS
- DFS

Features:

- Complete
- Memory efficient
- Finds shallowest solution

Process:

Depth 0 → Depth 1 → Depth 2 → ...

12. List out the ways to evaluate an algorithm's performance.

The performance of a search algorithm is evaluated using:

1. Completeness
Whether solution is guaranteed.
2. Optimality
Whether best solution is found.
3. Time Complexity
Amount of computation time required.
4. Space Complexity
Amount of memory required.

These measures help compare search algorithms.

13. Compare and Contrast BFS and DFS.

Feature	BFS	DFS
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Data Structure	Queue	Stack
Search Method	Level-wise	Depth-wise
Memory Usage	High	Low
Completeness	Yes	No
Optimal Solution	Yes	No

Conclusion:

BFS guarantees shortest path, while DFS uses less memory.

14. Define Bidirectional Search.

Bidirectional Search is a search strategy where two simultaneous searches are performed:

- One from initial state
- One from goal state

The searches meet in the middle to find solution faster.

Advantages:

- Reduces search space
- Faster execution

Disadvantage:

Requires knowledge of goal state.

15. Represent Taxi Driver Agent using PEAS Description.

PEAS stands for:

- Performance Measure
- Environment
- Actuators
- Sensors

Taxi Driver Agent PEAS:

Component	Description
Performance Measure	Safety, speed, comfort, profit
Environment	Roads, traffic, pedestrians, weather
Actuators	Steering, accelerator, brake
Sensors	GPS, camera, speedometer

Purpose:
PEAS helps design intelligent taxi driving systems.

CO-2

1. List out the types of Uninformed Search Strategies.

Uninformed search strategies are search techniques that use no heuristic information and explore the search space blindly.

Types of uninformed search strategies:

1. Breadth First Search (BFS)
2. Depth First Search (DFS)
3. Uniform Cost Search
4. Depth Limited Search
5. Iterative Deepening Search
6. Bidirectional Search

These algorithms use only problem definition to find solutions.

2. List out the types of Informed Search Strategies.

Informed search strategies use heuristic information to guide the search toward the goal efficiently.

Types of informed search strategies:

1. Greedy Best First Search
2. A* Search Algorithm
3. Hill Climbing Search
4. Beam Search
5. AO* Search

These algorithms reduce search space using heuristic functions.

3. Compare and Contrast BFS and DFS.

Feature	BFS	DFS
Data Structure	Queue	Stack
Search Method	Level-wise	Depth-wise
Memory Usage	High	Low
Completeness	Yes	No
Optimal Solution	Yes	No

Breadth First Search explores nodes level by level, while Depth First Search explores deeply before backtracking.

4. Define Bidirectional Search.

Bidirectional Search is a search strategy in which search is performed simultaneously from:

- Initial state
- Goal state

The two searches proceed toward each other and meet in the middle.

Advantages:

- Reduces search time
 - Reduces search space
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5. Define Global Maximum and Global Minimum.

Global Maximum:

The highest value among all possible solutions in a search space.

Global Minimum:

The lowest value among all possible solutions in a search space.

These terms are commonly used in optimization problems and search algorithms.

6. Define Local Search.

Local Search is a search technique that continuously moves from one state to a neighboring state to improve the solution.

Features:

- Uses current state only
- Does not maintain search tree
- Used in optimization problems

Example:

Hill Climbing Algorithm.

7. What are Ridges in the context of Search Algorithms?

Ridges are difficult regions in the search space where the search algorithm cannot move directly toward the optimal solution due to steep or narrow paths.

In hill climbing:

- The algorithm may oscillate
- Progress becomes slow

Ridges prevent efficient movement toward global optimum.

8. Identify the drawbacks of the Hill-Climbing Search technique.

Drawbacks of Hill Climbing:

1. Can get stuck at local maximum
2. Cannot guarantee optimal solution
3. May fail on plateaus
4. Difficult to handle ridges
5. No backtracking mechanism

Therefore, hill climbing may not always find the best solution.

9. Define Annealing Process.

Annealing is a process used in Simulated Annealing algorithm where the system gradually reduces temperature to reach an optimal solution.

Features:

- Initially accepts bad moves
- Gradually reduces randomness
- Helps escape local maxima

Inspired by metal cooling process in physics.

10. Define Evolutionary Algorithm.

An Evolutionary Algorithm is an optimization technique inspired by biological evolution.

It uses:

- Selection
- Mutation
- Crossover
- Reproduction

to generate better solutions over generations.

Examples:

- Genetic Algorithm
 - Evolution Strategies
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11. What is a Sensor-less Problem in Artificial Intelligence?

A Sensor-less Problem is a problem in which the agent has no information about the current state of the environment.

Features:

- No percepts available
- Agent must rely on action sequences
- High uncertainty

Example:

A robot moving in a dark room without sensors.

12. Define Exploration Problem.

An Exploration Problem is a problem where an agent must discover unknown parts of the environment while searching for goals.

Features:

- Environment initially unknown
- Agent learns while moving
- Common in robotics

Example:

Robot exploring a new building.

13. Define Online Search Problem.

An Online Search Problem is a search problem where the agent does not know the complete environment in advance.

The agent:

- Collects information during execution
- Decides actions step by step

Applications:

- Robotics
 - Navigation systems
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14. Define the process of Genetic Algorithm.

The Genetic Algorithm process involves the following steps:

1. Initialize population
2. Evaluate fitness
3. Select parents
4. Apply crossover
5. Apply mutation
6. Generate new population
7. Repeat until optimal solution found

It is inspired by natural selection and genetics.

15. What is a Stochastic Game?

A Stochastic Game is a game that includes randomness or uncertainty in outcomes.

Features:

- Actions may produce unpredictable results
- Uses probability

- Opponent and chance both affect outcome

Examples:

- Dice games
- Card games

These games are commonly used in AI decision-making problems.

CO-3

1. What is the importance of optimal decisions in game playing?

Optimal decisions help the player choose the best possible move to maximize winning chances and minimize losses.

Importance:

- Improves game strategy
- Helps achieve best outcome
- Avoids poor decisions

Optimal decision-making is important in AI games such as chess and tic-tac-toe.

2. Explain the concept of the Minimax algorithm in game playing.

Minimax is a decision-making algorithm used in two-player games.

It works by:

- MAX player tries to maximize score
- MIN player tries to minimize score

The algorithm explores the game tree and selects the optimal move assuming both players play intelligently.

Applications:

- Chess
- Checkers

- Tic-Tac-Toe
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3. What is the significance of Alpha-Beta Pruning in game search?

Alpha-Beta Pruning improves the efficiency of the Minimax algorithm by eliminating branches that cannot affect the final decision.

Significance:

- Reduces search space
- Saves computation time
- Improves search speed

It produces the same result as minimax with fewer node evaluations.

4. Differentiate between Minimax and Alpha-Beta Pruning.

Feature	Minimax	Alpha-Beta Pruning
Node Exploration	Explores all nodes	Prunes unnecessary nodes
Speed	Slower	Faster
Efficiency	Less efficient	More efficient
Result	Optimal	Optimal

Alpha-Beta Pruning is an optimized version of Minimax.

5. How does Alpha-Beta Pruning improve the efficiency of Minimax?

Alpha-Beta Pruning removes branches that cannot influence the final decision.

It:

- Avoids unnecessary calculations
- Reduces number of explored nodes
- Speeds up search process

Condition for pruning:

$$\alpha \geq \beta$$

Thus, the algorithm searches deeper in less time.

6. Define a Stochastic Game with an example.

A Stochastic Game is a game that includes randomness or uncertainty in outcomes.

Features:

- Uses probability
- Outcomes are unpredictable
- Includes chance events

Example:

- Dice games
- Card games

In these games, results depend on both player actions and random events.

7. How do chance nodes influence decision-making in a Stochastic Game?

Chance nodes represent random events in a stochastic game.

They influence decision-making by:

- Introducing uncertainty
- Affecting possible outcomes
- Requiring probability calculations

AI systems must consider expected outcomes before making decisions.

8. Explain the role of evaluation functions in game-playing algorithms.

Evaluation functions estimate the quality or desirability of a game state.

Role:

- Helps choose best move
- Evaluates board positions
- Guides search algorithms

Used in:

- Minimax
- Alpha-Beta Pruning

Example:

In chess, evaluation function measures piece strength and board control.

9. What are the advantages of using Alpha-Beta Pruning in two-player games?

Advantages of Alpha-Beta Pruning:

1. Reduces search space
2. Improves efficiency
3. Speeds up decision-making
4. Allows deeper search
5. Produces optimal solution

It makes game-playing AI faster and more practical.

10. How does randomness affect decision-making in Stochastic Games?

Randomness introduces uncertainty in game outcomes.

Effects:

- Decisions become probabilistic
- Exact prediction becomes difficult
- AI must calculate expected utility

Players must consider both strategy and chance events.

11. What is the difference between Deterministic and Stochastic Games?

Feature	Deterministic Game	Stochastic Game
Randomness	No	Yes
Outcome	Predictable	Uncertain

Example

Chess

Dice game

Deterministic games depend only on player actions, while stochastic games involve probability.

12. How does the horizon effect impact game-playing strategies?

Horizon effect occurs when an AI cannot see important future events because of limited search depth.

Effects:

- Poor decision making
- Delayed recognition of threats
- Incorrect move selection

It reduces the quality of game-playing strategies.

13. Explain the significance of game trees in decision-making.

A game tree represents all possible moves and game states.

Significance:

- Helps analyze future moves
- Supports optimal decision-making
- Used in minimax and alpha-beta pruning

Each node represents a game state and edges represent moves.

14. What is the role of utility functions in game-playing algorithms?

Utility functions assign numerical values to game outcomes.

Role:

- Measures success or failure
- Helps compare game states
- Guides optimal decision-making

Example:

Outcome	Utility Value
Win	+1
Draw	0
Loss	-1

15. How does pruning help in reducing the search space in game-playing?

Pruning removes branches that are not useful for final decision-making.

Benefits:

- Reduces explored nodes
- Saves time and memory
- Improves algorithm efficiency

Alpha-Beta Pruning is commonly used to reduce search space in AI games.

1. How can planning be represented? Provide an example.

Planning can be represented using:

- States

- Actions
- Goals

A planning problem includes:

- Initial state
- Goal state
- Sequence of actions

Example:

Initial State:

At(Home)

Goal State:

At(College)

Action:

Move(Home,College)

Thus, planning represents how actions transform one state into another.

2. What are the components of a planning system?

The components of a planning system are:

1. Initial State
Starting condition of the problem.
2. Goal State
Desired final condition.
3. Actions/Operators
Operations used to change states.
4. Planning Algorithm
Method used to generate plan.

These components help the agent achieve goals systematically.

3. Define a Planning Graph.

A Planning Graph is a layered graph structure used in AI planning to represent:

- States
- Actions
- Dependencies

It contains alternating levels of:

- Proposition nodes
- Action nodes

Planning graphs help in efficient plan generation.

4. List the conditions for a Mutex Relation.

Mutex (Mutual Exclusion) relation occurs when two actions or states cannot occur simultaneously.

Conditions for Mutex relation:

1. Inconsistent effects
2. Interference
3. Competing needs
4. Inconsistent support

Mutex relations are used in planning graphs.

5. Explain contingency planning.

Contingency Planning is a planning method used when outcomes are uncertain.

It prepares:

- Alternative plans
- Backup actions

Features:

- Handles unexpected situations
- Uses conditional actions

Example:

Robot choosing different paths depending on obstacle detection.

6. State the rule for Partial Order Planning.

Partial Order Planning follows the rule:

Only necessary ordering constraints should be imposed on actions.

Independent actions can occur in any order.

Advantages:

- Flexible planning
 - Supports parallel execution
-

7. What is the role of Hierarchical Task Network (HTN) planning?

HTN planning decomposes complex tasks into smaller subtasks.

Role of HTN planning:

- Simplifies problem solving
- Organizes planning hierarchically
- Improves efficiency

Example:

Travel planning divided into booking, packing, and transportation tasks.

8. Explain Generalization and Overfitting in learning.

Generalization:

Ability of a model to perform well on unseen data.

Overfitting:

Condition where model learns training data too closely and performs poorly on new data.

Generalization improves accuracy, while overfitting reduces model performance.

9. Provide examples of False Negative and False Positive hypotheses.

False Positive:

Model predicts positive incorrectly.

Example:

Healthy person diagnosed as diseased.

False Negative:

Model predicts negative incorrectly.

Example:

Diseased person diagnosed as healthy.

These errors occur in classification systems.

10. What is the entailment constraint in logic?

Entailment means:

A statement logically follows from another statement.

If knowledge base KB entails statement α :

$KB \models \alpha$

then α must be true whenever KB is true.

Used in logical reasoning and theorem proving.

11. List the difficulties in the Current Best Learning approach.

Difficulties:

1. Sensitive to noisy data
2. May converge slowly
3. Risk of local optimum
4. Computational complexity
5. Difficulty handling incomplete data

These issues affect learning efficiency.

12. Define Least Commitment Search.

Least Commitment Search is a planning strategy that delays decisions until necessary.

Features:

- Avoids unnecessary constraints
- Maintains flexibility
- Used in Partial Order Planning

It helps generate efficient plans.

13. What is Explanation-Based Learning (EBL)?

Explanation-Based Learning is a learning technique where the system learns by analyzing explanations of examples.

Features:

- Uses prior knowledge
- Generalizes from examples
- Improves future problem solving

Applications:

- Expert systems
 - Intelligent tutoring systems
-

14. Define Action Schema in planning.

An Action Schema is a generalized representation of an action in AI planning.

It includes:

- Action name
- Preconditions
- Effects

Example:

Move(x,y)

Precondition:

At(x)

Effect:

At(y)

Action schemas help represent reusable actions.

15. What is PDDL in AI planning?

PDDL stands for:

Planning Domain Definition Language.

It is a standard language used to describe:

- Planning problems
- Actions
- Goals
- States

PDDL is widely used in automated planning systems and AI research.

1. Define Language and Grammar.

Language:

Language is a system of communication using words, symbols, and rules to express ideas and information.

Grammar:

Grammar is a set of rules that governs the structure and formation of sentences in a language.

Grammar helps in:

- Sentence construction
 - Syntax checking
 - Meaningful communication
-

2. What is an N-Gram Model?

An N-Gram Model is a statistical language model used to predict the next word using previous words.

Types:

1. Unigram
2. Bigram
3. Trigram

Example:

"I drink"

Prediction:

tea

N-Gram models are widely used in NLP and text prediction.

3. Define Smoothing in Natural Language Processing.

Smoothing is a technique used to handle zero probability problems in language models.

It assigns small probabilities to unseen words or sequences.

Purpose:

- Improves prediction accuracy
- Prevents zero probability errors

Example:

Laplace Smoothing.

4. List the characteristics of Information Retrieval.

Characteristics of Information Retrieval:

1. Retrieves relevant information
2. Handles large document collections
3. Supports keyword searching
4. Uses indexing techniques
5. Improves search efficiency

Applications:

- Search engines
 - Digital libraries
-

5. What is Information Extraction?

Information Extraction is the process of automatically extracting useful information from unstructured text.

It identifies:

- Names
- Dates
- Locations
- Relationships

Applications:

- News analysis
 - Medical records
 - Web mining
-

6. Differentiate between Open Clause and Closed Clause.

Feature	Open Clause	Closed Clause
Variables	Contains variables	No variables
Example	$P(x)$	$P(a)$
Usage	General statements	Specific facts

Open clauses are not fully specified, while closed clauses contain fixed values.

7. Define Phrase Structure Grammar.

Phrase Structure Grammar is a grammar system used to describe sentence structure using production rules.

It represents:

- Noun phrases
- Verb phrases
- Sentence formation

Example:

$S \rightarrow NP + VP$

Used in Natural Language Processing and syntax analysis.

8. What is Chomsky Normal Form?

Chomsky Normal Form (CNF) is a simplified form of Context-Free Grammar where production rules are restricted to specific formats.

Rules:

1. $A \rightarrow BC$
2. $A \rightarrow a$

Where:

- A, B, C are non-terminals
- a is terminal

CNF is used in parsing algorithms.

9. Define Augmented Grammar.

Augmented Grammar is a grammar extended with additional information such as:

- Semantic rules
- Constraints
- Features

It improves language understanding and parsing accuracy.

Applications:

- NLP systems
 - Machine translation
-

10. What is Ambiguity? List its types.

Ambiguity occurs when a sentence has more than one meaning.

Types of ambiguity:

1. Lexical Ambiguity
2. Syntactic Ambiguity
3. Semantic Ambiguity

Example:

"I saw a man with a telescope."

Meaning can vary depending on interpretation.

11. Define Disambiguation in NLP.

Disambiguation is the process of identifying the correct meaning of an ambiguous word or sentence.

Purpose:

- Improves language understanding
- Reduces ambiguity

Example:

Determining whether "bank" means:

- River bank
 - Financial bank
-

12. List the applications of Natural Language Processing.

Applications of NLP:

1. Chatbots
2. Machine Translation
3. Speech Recognition

4. Sentiment Analysis
5. Text Summarization
6. Information Retrieval
7. Question Answering Systems

NLP enables computers to understand human language.

13. What are the different types of chatbots?

Types of chatbots:

1. Rule-Based Chatbots
2. AI-Based Chatbots
3. Hybrid Chatbots
4. Menu-Based Chatbots
5. Task-Oriented Chatbots
6. Conversational Chatbots

These chatbots are used in customer service and virtual assistance.

14. How are Machine Translation Systems implemented?

Machine Translation Systems are implemented using:

- Natural Language Processing
- Statistical Models
- Neural Networks
- Language Models

Steps involved:

1. Text analysis
2. Parsing
3. Translation
4. Sentence generation

Examples:

- Google Translate
- DeepL

Machine Translation converts text from one language to another automatically.